EXP 4 ADC

ADCexp2.m | ADCexp3.m | ADCexp4.m

/MATLAB Drive/ADCexp4.m

```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;         % Message frequency
fc = 10000;       % Carrier frequency
fLO = 14550;      % Local Oscillator (IF = 4550 Hz)

% Message Signal
m = cos(2*pi*fm*t);

% Carrier Signal
c = cos(2*pi*fc*t);

% AM Signal
am = (1 + m).*cos(2*pi*fc*t);

% Local Oscillator
LO = cos(2*pi*fLO*t);

% Frequency Mixing
mixed = am .* LO;

% IF Filter
[b,a] = butter(4,5000/(fs/2));
IF = filter(b,a,mixed);

% Envelope Detection
env = abs(hilbert(IF));

% Display
figure;

subplot(5,1,1);
plot(t,m);
title('Message Signal');
grid on;

subplot(5,1,2);
plot(t,c);
title('Carrier Signal');
grid on;

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46
47      subplot(5,1,3);
48      plot(t,am);
49      title('AM Signal');
50      grid on;
51
52      subplot(5,1,4);
53      plot(t,IF);
54      title('Intermediate Frequency (IF) Signal');
55      grid on;
56
57      subplot(5,1,5);
58      plot(t,env);
59      title('Recovered Signal');
60      grid on;
```

Figure 1

